

1. Definition

- $F_0 = 0, F_1 = 1$
- $F_n = F_{n-1} + F_{n-2}$ for $n \geq 2$

2. Combinatorial Formula

$$F_n = \sum_{k=0}^{\lfloor \frac{n-1}{2} \rfloor} \binom{n-k-1}{k}$$

3. Binet's Formula

$$F_n = \frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2} \right)^n - \frac{1}{\sqrt{5}} \left(\frac{1-\sqrt{5}}{2} \right)^n$$

4. Sum of First n Fibonacci Numbers

$$\sum_{i=1}^n F_i = F_{n+2} - 1$$

5. Sum of Odd-indexed Fibonacci Numbers

$$\sum_{i=0}^{n-1} F_{2i+1} = F_{2n}$$

6. Sum of Even-indexed Fibonacci Numbers

$$\sum_{i=1}^n F_{2i} = F_{2n+1} - 1$$

7. Sum of Squares

$$\sum_{i=1}^n F_i^2 = F_n F_{n+1}$$

8. Cassini's Identity

$$F_m F_{n+1} - F_{m-1} F_n = (-1)^n F_{m-n}$$
$$F_{2n} = F_{n+1}^2 - F_{n-1}^2 = F_n (F_{n+1} + F_{n-1})$$

9. Addition Formulas

$$F_m F_n + F_{m-1} F_{n-1} = F_{m+n-1}$$
$$F_m F_{n+1} + F_{m-1} F_n = F_{m+n}$$

10. Fibonacci Test

- A natural number n is Fibonacci iff $5n^2 + 4$ or $5n^2 - 4$ is a perfect square

11. Divisibility Property

- Every k -th Fibonacci number is a multiple of F_k

12. **GCD Property**

$$\gcd(F_m, F_n) = F_{\gcd(m, n)}$$

13. **Coprimality**

- Any three consecutive Fibonacci numbers are pairwise coprime

14. **Periodicity Modulo n**

- Fibonacci sequence modulo n is periodic with Pisano period $\leq 6n$